discover a partial marriage of quantum mechanics with general relativity (Chapter 12), so struggling to understand a black hole's core might help us to discover the full marriage; it might lead us to the full laws of quantum gravity. And perhaps the nature of the core will hold the keys to other mysteries of the Universe: There is a similarity between the "big crunch" implosion in which, eons hence, our Universe might die, and the implosion of the star that creates a black hole's core. By coming to grips with the one, we might learn about the other.

For thirty-five years physicists have pursued Wheeler's holy grail, but with only modest success. We do not yet know for certain what inhabits a hole's core, and the struggle to understand has not yet taught us with clarity the laws of quantum gravity. But we have learned much—not least that whatever is inside a black hole's core *is indeed* intimately connected with the laws of quantum gravity.

This chapter describes a few of the more interesting twists and turns in the quest for Wheeler's holy grail, and where the quest has led thus far.

The first, tentative answer to "What is inside a black hole?" Came from J. Robert Oppenheimer and Hartland Snyder, in their classic 1939 calculation of the implosion of a spherical star (Chapter 6). Although the answer was contained in the equations they published, Oppenheimer and Snyder chose not to discuss it. Perhaps they feared it would only add fuel to the controversy over their prediction that the imploding star "cuts itself off from the rest of the Universe" (that is, forms a black hole). Perhaps Oppenheimer's innate scientific conservatism, his unwillingness to speculate, kept them quiet. Whatever the reason, they said nothing. But their equations spoke.

After creating a black-hole horizon around itself, their equations said, the spherical star continues imploding, inexorably, to infinite density and zero volume, whereupon it creates and merges into a *spacetime singularity.*

A singularity is a region where—according to the laws of general relativity—the curvature of spacetime becomes infinitely large, and spacetime ceases to exist. Since tidal gravity is a manifestation of spacetime curvature (Chapter 2), a singularity is also a region of infinite tidal gravity, that is, a region where gravity stretches all objects infinitely along some directions and squeezes them infinitely along others.